Practice Assignment: Linear Regression on Placement Data

Course: PGD in Data Science with AI

Objective:

This assignment helps you understand how Simple Linear Regression works using a dataset of CGPA and placement package.

Dataset Description:

1. cgpa → Student CGPA
2. package → Placement package in LPA

Tasks:

1. Load the dataset using pandas.
2. Display the first 5 rows.
3. Check the shape of the dataset.
4. Display the column names.
5. Check whether there are any missing values.
6. Show basic statistical summary of the dataset.
7. Create a scatter plot between CGPA and Package.
8. Write 2–3 lines about what you observe from the graph.
9. Separate the dataset into X (cgpa) and y (package).
10. Split the data into training set and testing set.
11. Import and train the Linear Regression model.
12. Find slope (coefficient) and intercept.
13. Predict the package for CGPA = 8.5 and CGPA = 9.0.
14. Compare actual vs predicted values.
15. Plot the regression line on the scatter plot.
16. Label both axes properly and give the graph a title.
17. Write the regression equation in the form y = mx + b.
18. Explain the meaning of slope, intercept, and whether CGPA is a good predictor.